MFA final assignment

2026

In the final assignment of this course, you will apply dynamic and static MFA to a case study. This assignment tests your ability to perform the MFA approaches learned throughout the course with real-world data using Python programming. You are expected to formulate and conceptualize an appropriate research question, perform an MFA that builds upon and integrates multiple concepts and techniques introduced throughout the course, and interpret and discuss the results.

Your assignment is composed of several deliverables:

1. A poster
2. A bundle of python code, STAN files and data
3. Presentation of your poster in a poster session

Emphasis is placed on communicating your work concisely, thoughtfully, and clearly. Only these three deliverables will be evaluated (that is, no written reports etc.!) Check the grading criteria in this document to ensure that you got it all covered.

This is strictly an individual assignment.

Contents

1. Dates and times

The poster session will be held on Thursday, June 25, 9:30-16:00 in the Reuvenshal (main hall) of the Van Steenis building. If you want us to print your poster, it must be uploaded on Brightspace by Tuesday, June 23, at 10:00 a.m. Late submissions cannot be printed by us, and you will need to print them yourself.

The deadline to submit your poster in pdf format and the code & data bundle on Brightspace, is Friday, June 26, 23:59. You can continue revising any part of your project (model, results, poster design – basically everything) up to that deadline, so feel free to use the session's experience to change or modify your final work. This is similar to international academic conferences, where lots of posters and oral presentations are yet-unpublished works in progress that get improved through presentation.

This means that at the poster session, only the presentation components of the grade will be evaluated. The rest of the evaluation components will be graded using the contents we see in the Brightspace assignment at the end of the day: your FINAL poster pdf file, python code, STAN files, and accompanying data.

So regardless of how mature/finalized is the work that you show in the poster session, your presentation of the poster should still comprehensively showcase what you've done up to this point.

2. Case study

Your case study must include:

- At least one material in one end-use/product/sector/commodity
- A historical timeseries of inflows, outflows, and stocks, some of which are obtained from the modeling results.
- At least two future scenarios until at least 2050.
- At least one dynamic MFA model.
- At least two static MFA models that extend (i.e. describe the static snapshot with more detail) two timeframes from the dynamic MFA model

You are expected to define the system: location(s), sector(s), system boundaries, inclusion and exclusion of lifecycle stages, stock-driven or flow-driven model(s), aggregation or disaggregation of processes, temporal range and timesteps, etc.

How to choose a topic?

- Choose a topic that enables you to utilize and showcase what you've learned throughout the course. This is more important than novelty.
- If you can't find or create the necessary input data rather quickly, it's probably a sign that it's not a good topic for the objectives of this assignment. Stay focused on the modeling and interpretations, not on creating input data.
- Think about the story of the end-use, not only the story of the material. For example, glass in bottles have a very different life cycle than glass in cathedral windows, and your research design should address this.
- Find a good balance between the lifetime of the products/materials and the length of the examined period, in a way that makes the research non-trivial and calls for a dynamic MFA. For example:
    - Food has a shelf life of days to weeks, which calls for modeling timesteps of days or weeks to actually see any dynamics. Timesteps of years will just show throughput.
    - The urban mining of construction materials in road tunnels might not be interesting, because the average lifetime is so long that we won't see any interesting dynamics before 2050, no matter how extreme the scenarios are.

    - On the other hand, the long lifetimes of forever chemicals might be interesting to model exactly for the same reasons that urban mining of tunnels might not be interesting.
- Choosing a topic whose data exhibits common modeling challenges (like sudden jumps, initial stock data, etc.) is a great way to show how you deal with such challenges. But it's also ok to choose a dataset that doesn't lead to such challenges and then you can focus on other interesting things. You decide what to emphasize, and why – and we'd like to see justifications for such choices.
- Choosing model years and/or scenarios for creating a static model should be done to demonstrate

How to connect the dynamic and static models

- Choosing model years and/or scenarios for creating static models should be done to demonstrate the link between static and dynamic models.
- The choice of timeframes for the static models should illustrate some important differences in flows, transfer coefficients etc., which contribute to answering the research questions. If static models are closely comparable between years/scenarios, this may be an unexpected outcome that will need to be explained
- Students should reflect on the benefits of modelling dynamic and static MFA individually and together

3. Preparing your poster

Poster contents

Similar to presentations, reports, and journal articles, your poster should include the following contents:

- Introduction including background, motivation, objectives / aims / questions
- Methods and data, which also includes the MFA system definitions
- Results
- Discussion including interpretations, analysis, critical reflection, linkages to the introduction and linkage to the larger context.
- References

The poster must include **at least 3 figures**. Tip: start from choosing the headline figures that together convey your research, and build the poster's contents and design around them.

Sample posters

See Brightspace for several example posters. These posters didn't all necessarily get full marks (and their grading schemes were different), they're just samples for your inspiration! Note how some items are common, note the variabilities in styles and balances, note how different presenters chose to emphasize different aspects of their work, etc.

Technical notes

The poster can be in landscape or portrait orientation. You must leave some empty space (“bleed”) as a frame, in case the printer clips the sides. The poster will be printed in A1 size (594 x 841 mm). These have the same height-to-width ratio of a single A4 page. This means that you can design the poster on an A4 template in Powerpoint or other graphic design software of your choice.

To fill the entire A1 paper size, the printer will enlarge and stretch your contents. Make sure that the resolution of your graphics etc. is very high (i.e. high DPI settings) so that they don't get pixelated when enlarged. Without going into too many technical details, the two easy ways to reduce this risk are to zoom in a lot into your poster on your computer screen and check whether things still look sharp (good) or if the pixels are showing (bad), and to use vector graphics like SVG whenever possible because they stay sharp no matter what.

We will be able to print your posters a few days ahead of the June 25th poster session. We will send you details regarding printing of the posters in due time, including when the poster files need to be ready to be sent out to print. Please prepare your posters by this date, otherwise it will become your responsibility to print the poster on your own.

If you choose to print your own poster, simply bring it with you on Thursday morning (the 25th of June) directly to the session. We again emphasize that we encourage not printing your own poster because it can get expensive, and recommend you to submit your pdfs on Brightspace (the “optional: poster printing” assignment) on time and we’ll print them for you.

4. The poster session

The poster session will be held on Thursday, June 25, 9:30-16:00. Similar to poster sessions in academic conferences, you will stand next to your poster and use it to explain your work upon request to the audience which will go around the room.

The session will be split into two rounds, so that half of the students present their posters at any given time. When you’re not presenting, you will get tasks to look at other posters, so everyone must attend the entire session.

Feel free to invite your fellow students, friends, colleagues etc. to the poster session as audience - it's open to everyone.

5. Your code & data bundle

The code and data bundle should include the Python code .py file(s) or .ipynb notebook(s) and the input data in Excel tables or other formats that the python code can read, the STAN files for each static MFA assessed (.zmfa), and any other inputs. The poster, code and STAN files must fully describe the entire work in a stand-alone manner. Only the information that can be clearly and immediately inferred from these submitted files will be assessed, though in special cases the assessors may contact you for further clarifications.

This should be prepared so that when we run your submission on our computer we can fully recreate your results and trace your model, including all of the numbers that you use in your tables, figures, and text, and basically anything that you calculated yourself as part of this study. Check that your code runs on other machines before submission.

You are welcome and encouraged to reuse tidbits of codes from the MFA course's weekly assignments, answers, and in-class activities – this showcases that you learned to adapt MFA for your own usage. If you use any outside help writing your code (other students, AI/LLMs, help websites, repositories, etc.), you must explicitly state so and provide proper references. And it's still your responsibility to make sure you understand the code, confirm it does what it's supposed to do, and provide good documentation. Also note that AI tends to suggest generic and inefficient implementations, which is sometimes even incomplete. In any case, when asked, you should be able to explain each line of code and how it all works.

The code provided in the course is sufficient as a starting point for the modeling in the final assignment. If AI is used to supplement or improve the provided code, you need to make that clear, and you may be invited to explain your code to the instructors. If AI or any other source code is used to replace the provided code, you will be invited to explain your code to the instructors.

Place effort into making your code self-explanatory, give variables meaningful names (e.g. *inflow* , not *i*), and add comments to make it clear and easy to follow, and follow good practices. This should go beyond the weekly assignments' solutions we provide during the course – we kept those minimalistic on purpose for didactic reasons (to encourage you to explore), but this time your target audience is someone who didn't take this course.

Minimize calculations in Excel, do as much of the calculations and (pre)processing of your input data in Python to make it easy to track and replicate.

You don't have to provide the numerical modeling output results in an Excel file or as figures because the code should be able to create these results. Providing code for plotting figures is optional but recommended.

6. Further notes

- Contents will be checked with Turnitin.
- Minimum pass grade is 5.5 for the final assignment.
- A retake is possible, but the maximum grade to be obtained is then 6.

7. Grading criteria

1) **Motivation, literature review, research question, and objective**

The motivation for your research should be clear and relevant to the MFA course, and you should anchor it to the state of the art by linking it with relevant scientific publications. You are expected to formulate a research question that follows logically from the literature review and research objective.

- ≤5: Relation to course unclear and/or no publications cited; Research question and/or objective unclear, undefined, inappropriate or unrelated to MFA

- 6-7: Relation to course clear with a few publications cited, a link between the literature review and identified knowledge gap is provided; Research question follows knowledge gap and is related to MFA
- 8-9: Relation to course explained well, knowledge gap identified clearly, based in state-of-the-art academic literature with detailed referencing; Well-formulated research question, meets knowledge gap and research objective, and is related to MFA
- 10: Relation to course clear and referencing rich, to the point, adequate and clear, literature review leads to a knowledge gap definition that displays critical awareness of ongoing debates; Original and convincing research question firmly based on knowledge gap and research objective and is clearly related to MFA

2) Scenario formulation and set-up

Your scenarios should support the exploration of the research question and objective of the study. The scenario storylines and their logic should be clearly defined, explained, and justified, including in relation to each other. The future storylines should manifest in numbers that are appropriate in reflecting the storylines in a coherent and logical fashion.

- ≤5: The number of scenarios are partially or completely missing
- 6-7: The required number of scenarios are included. The scenarios support answering the research question and objective. The storylines and their conversion to numbers are adequate.
- 8-9: Well-formulated scenario storylines that clearly support answering the research question and study objective, with clear relations of similarities and differences between the scenarios. The scenarios are reflected well by numbers.
- 10: Exceptional scenarios firmly based on research objective, at the forefront of the field

3) Data sources & processing

You are expected to obtain and use citable data from reliable sources, correctly reference and describe them, describe the steps required for their processing and preparation (including assumptions, simplifications, proxies, pre-calculations, etc.) to become useful as data inputs.

- ≤5: Data sources or processing steps missing and/or inappropriate to address the research question
- 6-7: Data sources and processing steps appropriate to address the research question
- 8-9: Data sources and processing steps appropriate to address the research question, sources and processes are well referenced and described clearly
- 10: Data sources and processing steps appropriate to address the research question, sources and methods are well referenced and described, with a critical discussion on data requirements and alternative sources and some form of validation performed (i.e., comparison of data with external sources, or internal consistency check)

4) Methodology and modeling

You are expected to design dynamic and static MFA models and apply techniques and concepts learned in the course and correctly describe them.

- ≤5: Analytical method and/or model inappropriate to address the research question
- 6-7: Analytical method and/or model appropriate to address the research question
- 8-9: Analytical method and model appropriate to address the research question and described with clarity, with the help of a workflow diagram and/or equations
- 10: Advanced analytical method and model are appropriately applied to address the research question and described with clarity, with the help of a workflow diagram and/or equations that make the workflow clear and show the logic of the approach. Critical awareness of underlying assumptions shown.

5) Results

You are expected to report your main findings in the assignment. Tables and figures are required.

- ≤5: Results do not answer the research question
- 6-7: Results answer the research question
- 8-9: Results answer the research question with clarity and detail
- 10: A rich set of results that answer the research question with clarity and detail from multiple perspectives

6) Reflection on technical issues

You are expected to identify and deal with technical issues that arise from: data; modeling choices; model limitations, simplifications, and uncertainties; MFA troubleshooting issues such as negative inflows, initial stocks, sudden jumps, unbalanced flows; and so on.

- ≤5: No identification of potential issues such as listed above, or misunderstanding of the impacts of such issues.
- 6-7: Simple flagging of identified technical issues.
- 8-9: Identification of technical issues and discussion of their potential impacts on the results and interpretation of those results.
- 10: Detailed identification, analysis, discussion, and reflection of technical issues and their potential impacts on the results and interpretation of those results.

7) Analysis and discussion

You are expected to interpret your main findings in a manner that reflects on the work and its objective: What is the story? What do the results mean for your objectives (e.g. resource use, for sustainability, for supply chains, for circularity, for provisioning of services, for environmental impacts, etc.)? What are potential explanations for unexpected results? What are potential hurdles or gaps in interpreting and using the results? What more can be done? Tables and figures are helpful for this.

- ≤5: No analysis or discussion of the results, or analysis and discussion are irrelevant to the research question and objective, or analysis and discussion are generic and don't rely on the results
- 6-7: Adequate analysis, interpretation, and discussion of the results from one or two perspectives
- 8-9: Analysis, interpretation, and discussion of the results from several perspectives in clear connection to the knowledge gap and research question, including limitations to interpretations.
- 10: Analysis, interpretation, reflection, and discussion of the results from multiple in clear connection to the knowledge gap and research question, reflecting on open issues and challenges related to uncertainties and to the research process, and critically contrasting the findings with existing understanding revealed by the literature

8) **Scientific rigor and poster layout**

You are expected to make systematic and consistent use of terminology, units, and notation throughout the presentation. Your poster should show the research logically and clearly.

- ≤5: Fragmented, no or weak logical order in the poster's layout. Use of terminology, units, or notation frequently inconsistent with conventions
- 6-7: Straightforward layout is logical and readable. Adequate use of terminology, units, or notation consistent with conventions
- 8-9: Purposeful (i.e., with good reasons) design and layout, use of figures or tables clearly supporting the logical flow of the presentation of information. Good clarity, appealing visualizations are used purposefully to highlight the results. High level use of terminology, units, or notation consistent with conventions
- 10: Exceptional use of the poster format, clear and well-balanced use of figures, tables, text, etc. that support the logical flow of information. Outstanding clarity, appealing visualizations are used purposefully and effectively to highlight the reasoning and critical messages of the results. High level use of terminology, units, or notation consistent with conventions with exceptional care given to details (e.g., the choice of decimal cases relative to that of source data)

9) **Poster presentation**

You are expected to present the research within the time limit, in a logical order, and answer questions. Use the poster's contents to communicate the work.

- ≤5: No presentation, or fragmented, no or weak line of reasoning and lack of logical order, unable to answer questions
- 6-7: Straightforward presentation with line of reasoning and order is logical, able to answer questions.

- 8-9: Clear and streamlined presentation with a convincing line of reasoning, good use of the poster's contents, answers can develop to a discussion
- 10: Excellent reporting clarity, outstanding presentation highlighting the logic, reasoning and critical messages of the results, great use of the poster to support communication of the work. The questions and answers lead to an interesting discussion.

10) **Python code & STAN files**

The code and STAN files submitted should correspond to the poster's material, be designed to address the core research questions, and follow good programming/modelling practices. When asked, you should be able to explain each line of code, and each component of the static MFA model and how it works.

- ≤5: Connection with source data, processing, and results presented in the poster unclear, and/or the code/model doesn't work, and/or can't describe or explain how the code/model works.
- 6-7: The code and model work, connection with source data, processing, and results presented in the poster is clear. Can describe what each part of the code/model does.
- 8-9: The code and model work, connection with source data, processing, and results presented in the poster is clear, and the code has sectioning and comments for clarity. Can explain the code and model in detail.
- 10: The code and model is exemplary, clean, and efficient; connection with source data, processing, and results reported in the report clear; and the code is self-explanatory with sectioning and comments. Can explain the code and model in detail, reflecting on the logic behind modelling choices.

11) **Further assessment**

Points may be taken off if some required contents listed in these instructions are missing from the poster or code.

A modest addition to the grade may be given at the teachers' discretion for especially outstanding submissions, for example if contents significantly exceed the basic requirements in a meaningful way that contributes to the work (quality over quantity).

Good hunting!